

YAQIN CHEN

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RESEARCH INTERESTS

Generative Modeling and its Foundations: Deep Generative Models, Statistical Machine Learning, AI for Science

Sequential Decision Making and Optimization: Reinforcement Learning, Optimization Methods

EDUCATION

Northwestern University

Sep 2026 – Expected 2031

- Ph.D. in Industrial Engineering and Management Sciences (IEMS)

School of Mathematics (Zhuhai), Sun Yat-sen University

Sep 2022 – Jun 2026 (expected)

- B.S. in Information and Computing Science, GPA: 4.0/4.0 (1/60)

Department of Statistics, University of Wisconsin-Madison

Sep 2025 – May 2026

- Visiting International Student Program (Thematic Statistics Track)

RESEARCH EXPERIENCE

Department of Statistics, University of Wisconsin-Madison *Research Assistant*

Oct 2025 – Present

Undergraduate Thesis (in progress) Advisor: [Yiqiao Zhong](#)

- Understanding geometric structures of internal representations in coding and reasoning LLMs.
- Focusing on how representation geometry relates to generalization, robustness, and systematic reasoning behavior.

Decision, Risk and Operations Division, Columbia Business School *Research Assistant (remote)*

Mar 2025 – Present

Advisor: [Hongseok Namkoong](#)

- Developed the core scalable benchmarks for evaluating large language models in interactive, multi-turn environments.
- Benchmarked frontier LLM in-context learning across repeated interactions, identifying key adaptation challenges.
- Analyzed LLM failure modes via user simulation, providing key evidence of poor learning from experience.

iSURE Program, University of Notre Dame *Research Intern*

July 2025 – Dec 2025

Advisor: [Jianxun Wang](#) ([Sibley School of Mechanical Engineering, Cornell University](#))

- Researched the theoretical and mathematical unification of diffusion models and flow-matching generative frameworks.
- Adapted and benchmarked posterior sampling for flow-based conditional generation on scientific datasets (e.g., turbulence).
- Investigated training-free guidance techniques to accelerate sampling speed while ensuring generation accuracy.

Human Cyber Physical Intelligence Laboratory, Sun Yat-sen University *Research Assistant*

Feb 2024 – Feb 2025

Advisor: [Pengxu Wei](#), [Liang Lin](#) ([Fellow of IEEE/IAPR](#))

- Researched the impact of latent alignment and structural inductive biases in Variational Autoencoders (VAEs).
- Developed a statistical framework for efficient representation learning with Minimum Variance Unbiased Estimator.
- Devised novel autoencoder architectures that enforce statistical efficiency to improve representation learning quality.

ACADEMIC PAPERS

Benchmarking In-context Experiential Learning Through Repeated Product Recommendations [\[arXiv\]](#) [\[pdf\]](#)

Gilbert Yang*, [Yaqin Chen](#)*, Thomson Yen*, Hongseok Namkoong

Training-Free Guided Generation Techniques for Conditional Flow Matching on Scientific Datasets [\[arXiv\]](#) [\[pdf\]](#)

Meet Hemant Parikh, [Yaqin Chen](#), Jianxun Wang

When Local Rules Create Global Order: Self-Organized Representation Learning for Latent Diffusion Models [\[pdf\]](#)

Junrong Lian, Weijian Deng, Pengxu Wei, [Yaqin Chen](#), Qixiang Ye, Liang Lin

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

SELECTED PROJECTS

Development of a Martial Arts Teaching and Scoring System Based on YOLOv8

2024 Chinese Academy of Sciences Innovation and Practice Training Program For Undergraduate

Project Leader

Demo Pages

- Developed a real-time martial arts coaching system using YOLOv8 for pose estimation and movement analysis.
- Engineered an automated scoring engine based on video analysis to provide objective performance feedback.
- Integrated a Large Language Model (LLM) to generate personalized, multi-modal (text and voice) coaching.

AWARDS

Chinese National Scholarship (Top 0.2%)	2024
First-Class Excellent Student of SYSU (Top 5%, 1/151)	2023, 2024
MCM Modeling Contest Meritorious Award (7% of 11296)	2023

PROFESSIONAL SKILLS

Programming Languages: Python, C++, R, MATLAB

Languages: English (fluent), Chinese (native)